

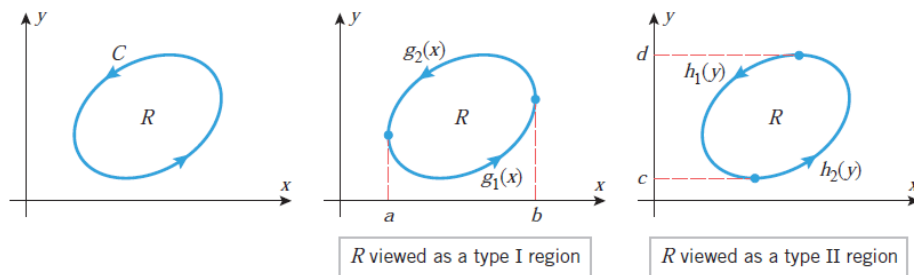
15.4 GREEN'S THEOREM

George Green (1793–1841) English mathematician and physicist. Green left school at an early age to work in his father's bakery and consequently had little early formal education. When his father opened a mill, the boy used the top room as a study in which he taught himself physics and mathematics from library books. In 1828 Green published his most important work, *An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism*. Although Green's Theorem appeared in that paper, the result went virtually unnoticed because of the small pressrun and local distribution. Following the death of his father in 1829, Green was urged by friends to seek a college education. In 1833,

after four years of self-study to close the gaps in his elementary education, Green was admitted to Caius College, Cambridge. He graduated four years later, but with a disappointing performance on his final examinations—possibly because he was more interested in his own research. After a succession of works on light and sound, he was named to be Perse Fellow at Caius College. Two years later he died. In 1845, four years after his death, his paper of 1828 was published and the theories developed therein by this obscure, self-taught baker's son helped pave the way to the modern theories of electricity and magnetism.

15.4.1 THEOREM (Green's Theorem) Let R be a simply connected plane region whose boundary is a simple, closed, piecewise smooth curve C oriented counterclockwise. If $f(x, y)$ and $g(x, y)$ are continuous and have continuous first partial derivatives on some open set containing R , then

$$\int_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA \quad (1)$$



► Figure 15.4.1

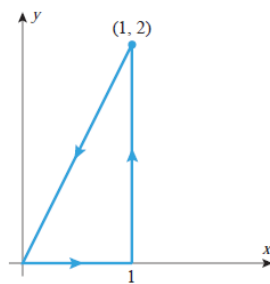
► **Example 1** Use Green's Theorem to evaluate

$$\int_C x^2 y \, dx + x \, dy$$

along the triangular path shown in Figure 15.4.3.

Solution. Since $f(x, y) = x^2 y$ and $g(x, y) = x$, it follows from (1) that

$$\begin{aligned} \int_C x^2 y \, dx + x \, dy &= \iint_R \left[\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(x^2 y) \right] dA = \int_0^1 \int_0^{2x} (1 - x^2) \, dy \, dx \\ &= \int_0^1 (2x - 2x^3) \, dx = \left[x^2 - \frac{x^4}{2} \right]_0^1 = \frac{1}{2} \end{aligned}$$



▲ Figure 15.4.3

Remark:

This agrees with the result obtained in Example 10 of Section 15.2, where we evaluated the line integral directly. Note how much simpler this solution is. ◀

■ **A NOTATION FOR LINE INTEGRALS AROUND SIMPLE CLOSED CURVES**

It is common practice to denote a line integral around a simple closed curve by an integral sign with a superimposed circle. With this notation Formula (1) would be written as

$$\oint_C f(x, y) \, dx + g(x, y) \, dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

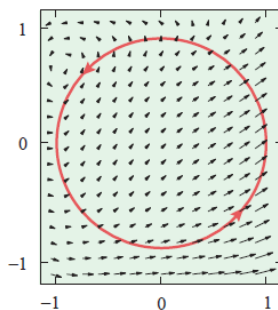
Sometimes a direction arrow is added to the circle to indicate whether the integration is clockwise or counterclockwise. Thus, if we wanted to emphasize the counterclockwise direction of integration required by Theorem 15.4.1, we could express (1) as

$$\oint_C f(x, y) \, dx + g(x, y) \, dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA \quad (5)$$

► **Example 2** Find the work done by the force field

$$\mathbf{F}(x, y) = (e^x - y^3)\mathbf{i} + (\cos y + x^3)\mathbf{j}$$

on a particle that travels once around the unit circle $x^2 + y^2 = 1$ in the counterclockwise direction (Figure 15.4.4).



▲ Figure 15.4.4

Solution. The work W performed by the field is

$$\begin{aligned} W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (e^x - y^3) dx + (\cos y + x^3) dy \\ &= \iint_R \left[\frac{\partial}{\partial x} (\cos y + x^3) - \frac{\partial}{\partial y} (e^x - y^3) \right] dA \\ &= \iint_R (3x^2 + 3y^2) dA = 3 \iint_R (x^2 + y^2) dA \\ &= 3 \int_0^{2\pi} \int_0^1 (r^2) r dr d\theta = \frac{3}{4} \int_0^{2\pi} d\theta = \frac{3\pi}{2} \quad \blacktriangleleft \end{aligned}$$

Green's Theorem

We converted to polar coordinates.

■ FINDING AREAS USING GREEN'S THEOREM

Green's Theorem leads to some useful new formulas for the area A of a region R that satisfies the conditions of the theorem. Two such formulas can be obtained as follows:

$$A = \iint_R dA = \oint_C x dy \quad \text{and} \quad A = \iint_R dA = \oint_C (-y) dx$$

Set $f(x, y) = 0$ and $g(x, y) = x$ in (1).

Set $f(x, y) = -y$ and $g(x, y) = 0$ in (1).

A third formula can be obtained by adding these two equations together. Thus, we have the following three formulas that express the area A of a region R in terms of line integrals around the boundary:

$$A = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C -y \, dx + x \, dy \quad (6)$$

► **Example 3** Use a line integral to find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Solution. The ellipse, with counterclockwise orientation, can be represented parametrically by

$$x = a \cos t, \quad y = b \sin t \quad (0 \leq t \leq 2\pi)$$

If we denote this curve by C , then from the third formula in (6) the area A enclosed by the ellipse is

$$\begin{aligned} A &= \frac{1}{2} \oint_C -y \, dx + x \, dy \\ &= \frac{1}{2} \int_0^{2\pi} [(-b \sin t)(-a \sin t) + (a \cos t)(b \cos t)] \, dt \\ &= \frac{1}{2} ab \int_0^{2\pi} (\sin^2 t + \cos^2 t) \, dt = \frac{1}{2} ab \int_0^{2\pi} dt = \pi ab \quad \blacktriangleleft \end{aligned}$$

■ GREEN'S THEOREM FOR MULTIPLY CONNECTED REGIONS

$$\begin{aligned} \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA &= \iint_{R'} \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA + \iint_{R''} \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA \\ &= \oint_{\substack{\text{Boundary} \\ \text{of } R'}} f(x, y) \, dx + g(x, y) \, dy + \oint_{\substack{\text{Boundary} \\ \text{of } R''}} f(x, y) \, dx + g(x, y) \, dy \end{aligned}$$

However, the two line integrals are taken in opposite directions along the cuts, and hence cancel there, leaving only the contributions along C_1 and C_2 . Thus,

$$\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA = \oint_{C_1} f(x, y) dx + g(x, y) dy + \oint_{C_2} f(x, y) dx + g(x, y) dy \quad (7)$$

which is an extension of Green's Theorem to a multiply connected region with one hole. Observe that the integral around the outer boundary is taken counterclockwise and the integral around the hole is taken clockwise. More generally, if R is a multiply connected region with n holes, then the analog of (7) involves a sum of $n + 1$ integrals, one taken counterclockwise around the outer boundary of R and the rest taken clockwise around the holes.

► **Example 4** Evaluate the integral

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2}$$

if C is a piecewise smooth simple closed curve oriented counterclockwise such that (a) C does not enclose the origin and (b) C encloses the origin.

Solution (a). Let

$$f(x, y) = -\frac{y}{x^2 + y^2}, \quad g(x, y) = \frac{x}{x^2 + y^2} \quad (8)$$

so that

$$\frac{\partial g}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial f}{\partial y}$$

if x and y are not both zero. Thus, if C does not enclose the origin, we have

$$\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} = 0 \quad (9)$$

on the simply connected region enclosed by C , and hence the given integral is zero by Green's Theorem.

Solution (b). Unlike the situation in part (a), we cannot apply Green's Theorem directly because the functions $f(x, y)$ and $g(x, y)$ in (8) are discontinuous at the origin. Our problems are further compounded by the fact that we do not have a specific curve C that we can parametrize to evaluate the integral. Our strategy for circumventing these problems will be to replace C with a specific curve that produces the same value for the integral and then use that curve for the evaluation. To obtain such a curve, we will apply Green's Theorem for multiply connected regions to a region that does not contain the origin. For this purpose we construct a circle C_a with clockwise orientation, centered at the origin, and with sufficiently small radius a that it lies inside the region enclosed by C (Figure 15.4.6). This creates a multiply connected region R whose boundary curves C and C_a have the orientations required by Formula (7) and such that within R the functions $f(x, y)$ and $g(x, y)$ in (8) satisfy the hypotheses of Green's Theorem (the origin does not belong to R). Thus, it follows from (7) and (9) that

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} + \oint_{C_a} \frac{-y dx + x dy}{x^2 + y^2} = \iint_R 0 dA = 0$$

It follows from this equation that

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = - \oint_{C_a} \frac{-y dx + x dy}{x^2 + y^2}$$

which we can rewrite as

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \oint_{-C_a} \frac{-y dx + x dy}{x^2 + y^2}$$

Reversing the orientation of C_a reverses the sign of the integral.

But C_a has clockwise orientation, so $-C_a$ has counterclockwise orientation. Thus, we have shown that the original integral can be evaluated by integrating counterclockwise around a circle of radius a that is centered at the origin and lies within the region enclosed by C . Such a circle can be expressed parametrically as $x = a \cos t$, $y = a \sin t$ ($0 \leq t \leq 2\pi$); and hence

$$\begin{aligned} \oint_C \frac{-y dx + x dy}{x^2 + y^2} &= \int_0^{2\pi} \frac{(-a \sin t)(-a \sin t) dt + (a \cos t)(a \cos t) dt}{(a \cos t)^2 + (a \sin t)^2} \\ &= \int_0^{2\pi} 1 dt = 2\pi \quad \blacktriangleleft \end{aligned}$$

Practice Problem:

Chapter-15.4:

Problems: 2, 4, 5, 6, 7-13.

Thank you for your attention!